

SIMATS Online Programming Lab

PYTHON PROGRAMMING

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```
46 lines = []
47 lines.append("Advanced Text Statistics Analyser")
48 lines.append(f"Characters (including spaces): {stats['characters_including_spaces']}")
49 lines.append(f"Characters (excluding spaces): {stats['characters_excluding_spaces']}")
50 lines.append(f"Words: {stats['words']}")
51 lines.append(f"Sentences: {stats['sentences']}")
52 lines.append(f"Unique words: {stats['unique_words']}")
53 lines.append(f"Average word length: {stats['avg_word_length']:.2f}")
54 lines.append(f"Average words per sentence: {stats['avg_words_per_sentence']:.2f}")
55 lines.append(f"Non-stopword words: {stats['non_stop_word_count']}")
56 lines.append("Top 10 words:")
57 for w, c in stats["top_10_words"]:
58     lines.append(f" {w}: {c}")
59 return "\n".join(lines)
60
61 if __name__ == "__main__":
62     user_input = input("Enter text: ")
63     stats = analyze_text(user_input)
64     print()
65     print(format_report(stats))
```

Hello World

OUTPUT

```
Enter text:
Advanced Text Statistics Analyser
Characters (including spaces): 11
Characters (excluding spaces): 10
Words: 2
Sentences: 1
Unique words: 2
Average word length: 5.00
Average words per sentence: 2.00
```

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PROGRAM EDITOR

Python

Run

Save

```
1 def compress_rle(s):
2     if not s:
3         return ""
4     result = []
5     count = 1
6     for i in range(1, len(s)):
7         if s[i] == s[i-1]:
8             count += 1
9         else:
10            result.append(s[i-1] + (str(count) if count > 1 else ""))
11            count = 1
12    result.append(s[-1] + (str(count) if count > 1 else ""))
13    return "".join(result)
14
15 s = input().strip()
16 print(compress_rle(s))
```

INPUT

Hello World

OUTPUT

Hel2o World

PROGRAM EDITOR

Python

Run

Save

```
1 s = input("Enter a string: ").lower()
2
3 vowels = set("aeiou")
4 vowel_count = 0
5 consonant_count = 0
6 other_count = 0
7
8 for ch in s:
9     if ch.isalpha():
10         if ch in vowels:
11             vowel_count += 1
12         else:
13             consonant_count += 1
14     else:
15         other_count += 1
16
17 print("Vowels:", vowel_count)
18 print("Consonants:", consonant_count)
19 print("Others:", other_count)
```

INPUT

Hello World

OUTPUT

```
Enter a string: Vowels: 3
Consonants: 7
Others: 1
```

```
1 s = input("Enter a string: ")
2 freq = {}
3
4 for ch in s:
5     freq[ch] = freq.get(ch, 0) + 1
6
7 for ch in sorted(freq):
8     print(f"{ch}: {freq[ch]}")
```

hello world

UTPUT

```
Enter a string:  : 1
d: 1
e: 1
h: 1
l: 3
o: 2
r: 1
w: 1
```